

- Question Analysis -

Masud $\rightarrow 6 \rightarrow 1$ set
 $8 \rightarrow 01$ set
 $7+9 \rightarrow 1$ set

Masud sir-

1. Define cooperating process. Develop a model of a system consisting of cooperating sequential process or thread with the producer-consumer problem. (4)
2. What is critical section problem? Explain the requirement which must be satisfied to solve the critical section problem. *
3. Explain the strategy, high level synchronization, construct monitor type, for dealing with such error. $\{(2+3) \rightarrow 7\}$
4. What is spinlock? state the effect of liveness failure in the process execution life cycle. (3) *
5. Write the formal definition of deadlock in multithreaded application. Point out the necessary conditions which hold simultaneously in system for deadlock situation. (4) * *
6. Illustrate and explain the resource allocation graph, resource-allocation graph with dead lock and resource allocation graph with cycle but no dead lock * *
7. How to detect dead lock in system? Illustrate and explain the data structure which must be maintained, process and example of deadlock detection algorithm with several instance of each resource type. (5)
8. Formulate reader-writer problem as classical synchronization problem. discuss the solution of the problem using binary and mutex semaphore (4)

9. Define and demonstrate the dining-philosophers problem as a synchronization problem. State the semaphore and monitor solutions of dining-philosophers problem. (4) *
10. Describe a mechanism for enforcing ~~an~~ memory protection in order to prevent a program from modifying the memory associated with other program. (3)
11. Compare and contrast monolithic, layered, microkernel, modular and hybrid strategies for designing operating system. (3)
12. What are the three main purposes of an operating system? How does the kernel mode and user mode function as a rudimentary form of protection? $\rightarrow$ [19-20]
13. How to arise data inconsistency in concurrent access? (2+3) *
14. Mention the limitation of Peterson's solution for addressing the ~~are~~ critical section problem. What are mutex ~~lock~~ locks? State the solution to critical section problem using mutex ~~lock~~ lock. (7) *
15. Describe the data structure which must be maintained to implement the banker's algorithm for deadlock avoidance. How to detect deadlock in a system with only a single instance of each resource type? *

16. for improving the utilization of the CPU and the speed of computer's response to its users, list various way to manage memory in operating system. Distinguish logical and physical address and mention the function of the memory management unit, MMU for generating address. (4)

17. What is paging? Explain and illustrate the paging model of logical and physical memory and paging example for a 32-byte memory with 4-byte pages. (5)

↳ [18-19] 5 question repeated

16. Why synchronization tools are needed to ensure concurrent access? How to solve the critical section problem using Peterson's solution. (7)

17. Define liveness in process execution life cycle. (2)

18. What are the side effect of preventing dead lock? (1st half of 15) 6

↳ [17-18] 4 question repeated